No Trading Strategy Can Win on Every Price Path: Computability, Randomness, and the Limits of Universal Trading

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Every universal-trading claim pairs a trader with a market—a path, generator, or law. For any total deterministic computable trader, its code yields a fixed computable countermarket with proportional price moves opposing its positions; hence no such trader wins on every computable path. Gold-style learning cannot identify every computable binary market rule from history. Separately, Turing-universal generators make certification of unbounded-future events undecidable; Busy-Beaver growth defeats every computable description-size waiting schedule. A passive Martin-Löf-random record is incompressible and prevents effective test-capital processes from becoming unbounded; no-arbitrage supplies a distinct financial boundary. Together these results form an expository taxonomy: repeatable success requires a market restriction, benchmark, risk premium, or informational advantage, while time reversal supplies a simple stress test.

I. INTRODUCTION

Throughout this paper the elementary configuration has two sides: a *trader* and a *market*. At each date the trader converts observed history into a position; the market supplies the next price. Depending on the question, the market side is a realized path, a path-generating program, a probability law, or a price process constrained by no-arbitrage. Holding the trader fixed while changing the admitted market side reveals which universal claim is at issue.

A strategy is *universal* here only if, after discounting by a specified numeraire, it produces strictly positive terminal gain from zero initial capital on every admissible trajectory. Zero cost, self-financing, admissibility, and discounted gain distinguish this claim from positive wealth obtained by investing positive capital. Its quantifier order is

$$\exists\,\sigma\;\forall\,\mathcal{E}: \qquad \sigma \text{ wins against } \mathcal{E}, \tag{1}$$

whereas diagonalization establishes

$$\forall\,\sigma\;\exists\,\mathcal{E}_\sigma: \qquad \sigma \text{ does not win against } \mathcal{E}_\sigma. \tag{2}$$

For a fixed total deterministic computable trader, its code can be embedded in a total program that recursively chooses the next bounded proportional price move opposite to the trader's position. The resulting countermarket is one fixed computable path; any finite prefix can be generated offline. A trader that never takes a position earns zero, which already refutes a guarantee of strict profit. This is the trading counterpart of the next-bit construction $x_0 = 1 - F(\varnothing)$ and $x_t = 1 - F(x_0, \ldots, x_{t-1})$ for $t \geq 1$: it defeats the fixed proposed rule, not every rule on one exceptional sequence.

Three complementary limits use different market information. Gold's learner sees an accumulating history and must eventually stabilize on a correct program; no effective learner identifies every binary total recursive time function, although restricted classes such as the primitive-recursive time functions are identifiable in the limit [1, 2]. A source-code procedure that settled every future event of Turing-universal generators would instead decide the Halting Problem, while Busy-Beaver growth excludes a computable worst-case waiting horizon. Finally, relative to a specified computable measure, a Martin-Löf-random record escapes every effective null test, has incompressible prefixes, and prevents effective nonnegative test-capital from becoming unbounded. No-arbitrage supplies a separate pricing-measure boundary.

Repeatable practical success must therefore identify structure absent from the unrestricted market class: persistence, mean reversion, a risk premium, a benchmark, lawful information, or another distributional asymmetry. Large, crowded, or public strategies may also change the distribution through price impact and adaptation. This falsifiable restriction on the market side is the operational meaning of "insight" used below.

Table I is the paper's principal contribution: a taxonomy that keeps distinct several statements commonly summarized as "no universal strategy." The diagonal countermarket and its proofs are expository reformulations rather than a claim of a newly discovered impossibility theorem. The analysis connects this quantifier map to the literature on unpredictable prices [3–5], Gold learning, computability and randomness, no-arbitrage and NFL, time-reversal stress testing, the Wheel, Cover universality, and conditional FAPP strategies.

II. THE TRADER AND THE MARKET: COMPUTABILITY AND RANDOMNESS

We now formalize the trader–market pair. The trader is a program M_σ mapping each observed finite price history to a position; the market side supplies the price history. Four specifications lead to four different bound-

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TABLE I. Distinct claims organized by the trader–market configuration. They are complementary, not interchangeable.

Mechanism	Market side	Formal scope	Conclusion
Diagonal countermarket	Fixed total program constructed from the fixed trader's code	Pathwise worst case: $\forall\sigma\,\exists P_\sigma$	No total deterministic computable trader wins on every computable price path
Gold identification in the limit	Accumulating computable history, or interaction with a black box	Success is eventual stabilization on a correct program, without a known stopping date	No effective learner identifies every computable binary time function; restricted classes can be identified
Halting reduction and Busy Beaver	Total program or generator, under an explicit promise	Individual encoded computation over an unbounded horizon	No general computable profit or market-event decider, or computable description-size waiting horizon
Algorithmic randomness	Passive binary record under the computable fair-coin reference law	Each Martin-Löf-random path; such paths have measure one	No constructive supermartingale, hence no computable fair-game martingale, becomes unbounded
No-arbitrage	Stochastic price process admitting a coherent pricing measure	Almost-sure payoff conditions under an ELMM	No admissible zero-cost classical arbitrage
Cover universality	Passive bounded sequence and a stated benchmark class	Pathwise, but benchmark-relative and asymptotic	Vanishing per-period regret, not guaranteed positive profit

aries: a fixed computable countermarket tailored to a trader defeats its claim of pathwise universality; Gold's history protocol precludes identifying every computable rule in the limit; a computationally universal class of market generators makes uniform certification of unbounded future events undecidable; and a passive algorithmically random record defeats effective fair betting.

A. A Computable Countermarket for Every Trader

Let M_σ be a *total* computable map from every finite computable price history H_t to a rational position w_t. Totality models the practical requirement that a deployed algorithm return an action before a deadline. Rational outputs model finite-precision orders and, crucially, make comparison with zero decidable. Let $\mathcal{P}^+_{\text{comp}}$ denote the positive rational-valued computable price paths.

Theorem 1 (Computable Diagonal Countermarket)**.** *For every total deterministic computable strategy M_σ, there exists a fixed path $P^{M_\sigma} \in \mathcal{P}^+_{\text{comp}}$ on which its self-financing gain*

$$G_T = \sum_{t=0}^{T-1} w_t(P_{t+1} - P_t)$$

is non-positive at every finite horizon T. It is strictly negative after the strategy has taken a nonzero position at least once.

Proof. Choose rational $P_0 > 0$ and rational $0 < \epsilon < 1$. Recursively compute $w_t = M_\sigma(P_0, \ldots, P_t)$ and set

$$P_{t+1} = \begin{cases} P_t(1-\epsilon), & w_t > 0, \\ P_t(1+\epsilon), & w_t < 0, \\ P_t, & w_t = 0. \end{cases} \tag{3}$$

Because M_σ is total and rational-valued, this recursion is a total program for one fixed infinite price path. Its prices remain positive and computable. In each period, $w_t(P_{t+1} - P_t) \le 0$, with strict inequality when $w_t \ne 0$. Summing proves the claim at every finite horizon. ☐

Remark 1 (Logical and Economic Scope of the Countermarket). Theorem 1 uses the trader's *code*, not a physical market's observation of a deployed order. Once M_σ is fixed, the countermarket program is fixed and can generate any desired finite prefix offline. It is therefore a lawful, deterministic, computable counterexample, although not necessarily an efficient one computationally. If the market side is regarded as an opponent whose objective is to prevent this trader's gain, the rule is rational relative to that objective. It is not thereby shown to be an economic equilibrium or an arbitrage-free price process, and another trader may profit on it. Such a trader has, in turn, its own computable countermarket.

This is a direct strategy-relative diagonal construction. The proposed universal trader is used to compute an object outside its success set: long is paired with down, short with up, and neutrality with no price change. Just as the complement sequence defeats a purported predictor of every computable sequence, P^{M_σ} defeats the purported trader on every horizon. Enumeration of all programs is unnecessary because the universal claim itself supplies the program to be diagonalized against. The quantifiers are $\forall M_\sigma\, \exists P^{M_\sigma}$. There need not be one path on which every strategy loses; for example, permanently long and permanently short positions cannot both lose on the same one-step price move.

B. Learning a Market Rule from History: Gold's Limit

A trader normally receives neither the market generator's source code nor a proof of its law. The trader sees an expanding record $x_0, x_1, \ldots$ and must infer from each finite prefix how it will continue. Gold's function-identification model formalizes exactly this information flow. At time n, an effective learner L maps $(x_0, \ldots, x_n)$ to the index of a Turing machine. It *identifies x in the limit* if, from some finite date onward, every guess is the same index and that machine computes the whole sequence. The learner need not know when its last change of mind has occurred [1, 2].

Let REC_2 denote the total computable binary sequences. Gold's negative theorem for binary-valued time functions has the trader–market quantifier order, with L ranging over effective learners,

$$\begin{aligned} &\forall L\ \exists x \in \mathrm{REC}_2 : \\ &\quad L(x_0, \ldots, x_n) \text{ does not stabilize} \\ &\quad \text{on a program for } x. \end{aligned} \tag{4}$$

The adverse history depends on the learner. This is closely analogous to $\forall M_\sigma\, \exists P^{M_\sigma}$, although Gold's failure criterion is failure to converge to a fixed correct program name, not financial loss.

The corresponding claim about literal next-move forecasting is sharper and has the short direct diagonal proof already used above. If a total computable predictor F outputs one bit from every finite binary history, the recursive definition

$$x_0 = 1 - F(\varnothing), \qquad x_t = 1 - F(x_0, \ldots, x_{t-1}) \quad (t \geq 1)$$

produces a computable market record on which F is wrong at every date. Thus the proposed statement—every total next-move predictor has a computable countermarket—is correct, but it is this direct prediction diagonalization, not the literal statement of Gold's identification theorem. Nonconvergence of program names alone does not count the learner's next-step errors.

The opposite-bit construction also appears explicitly in Legg's analysis of universal prediction [6]. The qualifier *computable* is essential. Solomonoff induction defines an uncomputable universal predictive ideal; for computable probabilistic environments its conditional predictions converge, with quantitative error bounds developed by Hutter [7–10]. It therefore evades, rather than contradicts, the diagonal result: it is not a total executable trading rule, and predictive convergence is not Gold's requirement of stabilizing on one exact generating program. For probability-valued forecasts, V'yugin gives a further, distinct impossibility: a randomized construction can make any partial computable forecaster fail calibration with probability arbitrarily close to one [11]. Calibration, next-bit error, and identification in the limit are three different criteria.

Gold's results also show why the restriction on the market side matters. Every effectively enumerable class of total functions is identifiable in the limit; in particular, primitive-recursive time functions are. In Gold's interactive black-box model, where the eventual hypothesis must reproduce future input–output behavior, finite-state boxes are weakly identifiable in the limit, whereas the class of primitive-recursive black boxes is not [1]. These are different object classes, not contradictory claims. Gold's 1967 language model further makes explicit that learnability depends on the admitted class, the presentation of the evidence, and the allowed names [2]. For trading, a forecast can therefore be justified only relative to a specified restriction on possible markets.

C. Universal Computation and the Halting Barrier

Gold's learner works from observations and may revise its hypothesis indefinitely. The next results ask a separate, stronger operational question: can one terminating program inspect arbitrary encoded traders or market generators and correctly decide their unbounded future behavior? The preceding diagonal construction already disproves a trader that wins on every computable path and does not invoke the Halting Problem. The source-code event-certification question does.

Theorem 2 (Undecidability of Eventual Profit)**.** *Fix the positive computable price path $P_t = t + 1$, $t \in \mathbb{N}$. There is no algorithm which, under the promise that its input code computes a total trading program M_σ, correctly decides for every such code whether*

$$\exists T \geq 1 : \qquad G_T(M_\sigma; P) = \sum_{t=0}^{T-1} w_t (P_{t+1} - P_t) > 0.$$

Proof. Assume such a profit certifier exists. Given an arbitrary Turing machine A and input x, construct a trading program $M_{A,x}$ as follows. At date t, simulate $A(x)$ for $t+1$ steps. If it has halted within those steps, set $w_t = 1$; otherwise set $w_t = 0$. Every date requires only a finite simulation, so $M_{A,x}$ is total. Along $P_t = t+1$, every long position earns one unit in the next period. Hence $G_T > 0$ for some T if and only if $A(x)$ eventually halts. A certifier for eventual trading profit would therefore decide the Halting Problem, which is impossible [12]. □

The reduction can also be placed directly on the market side, which is the form most closely connected to source-code event certification.

Theorem 3 (Undecidability of a Market Event)**.** *There is no algorithm which, under the promise that its input* *code computes a total positive-price generator G, correctly decides for every such code whether its generated path P^G ever satisfies $P_t^G > P_0^G$.*

Proof. Given a Turing machine A and input x, define the total generator by $P_0^{A,x} = 1$ and, for $t \geq 1$,

$$P_t^{A,x} = 1 + \mathbf{1}\{A(x) \text{ halts within } t \text{ steps}\}, \qquad t \geq 1. \quad (5)$$

Each quoted price requires only a finite t-step simulation, so the generator is total and computable. Its initial price is one, and it ever quotes a higher price if and only if $A(x)$ halts. A universal decider for the stated market event would therefore decide the Halting Problem. □

Theorem 3 makes "universal computation" precise: the admissible generators can emulate an arbitrary machine and expose its outcome through a price event. No algorithm can then certify every future event of every such market.

This is not Gold's theorem. The reduction receives source code and must halt with a yes-or-no answer; Gold's learner receives accumulating data and need only stabilize eventually. Indeed, the paths in Eq. (5) are identifiable in the limit: guess the flat path, then switch permanently if the price becomes two. No finite date, however, certifies that the initial guess is final. Thus halting excludes uniform certification, whereas Gold excludes identification in the limit for the class of all binary total recursive time functions. Restricted model classes may still be learnable [13–17].

a. Busy-Beaver horizons: finite glimpses and noncomputable waiting Fix a step-counted universal machine V with an effective compiler that adds only fixed overhead for a given program schema, and define the runtime Busy Beaver function by

$$\mathrm{BB}(n) = \max\{\mathrm{time}_V(p) : |p| \leq n \text{ and } V(p) \text{ halts}\}, \quad (6)$$

with value zero if the set is empty. The input n is a program-description budget, not a date; $\mathrm{BB}(n)$ is a runtime, which becomes a possible trading date when the computation is embedded in a market generator. A small increase in rule length may therefore require waiting beyond every computably prescribed scale.

Each $\mathrm{BB}(n)$ is finite, and selected low values can sometimes be proved by exhaustive simulation plus proofs that the remaining programs do not halt. This is not uniform "computability for small arguments": any finite table can be hard-coded, and its apparent pattern licenses no extrapolation. If a total computable h upper-bounded BB, simulating every program of length at most n for $h(n)$ steps would decide halting. Eventual domination uses a further program-schema argument. For each n, a program of $O(\log n)$ bits can encode n, compute $h(n)$, execute more than $h(n)$ dummy steps, and halt. For all sufficiently large n, its compiled V-code has length at most n, so $\mathrm{BB}(n) > h(n)$. Thus, up to the fixed machine convention, BB eventually exceeds every total computable function [18, 19].

Calude, Dinneen, and Shu illustrate the finite/infinite divide. For one explicit universal self-delimiting register machine, computation and mathematical proof settled halting through 84-bit programs and certified 64 initial bits of its halting probability Ω_U. They did not compute Busy Beaver values, and the work needed to recover the relevant halting programs from longer supplied prefixes of Ω_U outruns every computable bound. Their result is a formidable finite glimpse, not a scalable computation of an uncomputable object [20].

Applied to Eq. (5), a short generator can quote $P_t = 1$ for a Busy-Beaver-scale interval and then quote $P_t = 2$. Across encoded programs of length at most n, the last halt can occur at $\mathrm{BB}(n)$, up to fixed wrapper overhead. Knowing that horizon would make continued silence a certificate of no later move. Equivalently, with TotComp denoting total computable size-dependent schedules,

$$\begin{aligned} \forall h \in \mathrm{TotComp}\ \exists p: \quad & V(p) \text{ halts}, \\ & \mathrm{time}_V(p) > h(|p|). \end{aligned} \quad (7)$$

If c is the wrapper's fixed description-length overhead, it is absorbed by the computable envelope $H(n) = \max_{k \leq n+c} h(k)$. Every quote at a specified finite date remains computable by bounded simulation; the generator neither computes Busy Beaver nor selects its maximizer uniformly. Noncomputability lies in certification across the family. Each realized path is deterministic and computable, hence not Martin-Löf random under fair-coin coding.

The Taylor comparison is qualitative, not a race between steep curves. For an effectively specified analytic law f,

$$\begin{aligned} f(t_0 + \Delta) &= f(t_0) + f'(t_0)\Delta + R_2(\Delta), \\ R_2(\Delta) &= O(\Delta^2), \end{aligned} \quad (8)$$

under the usual local smoothness and remainder conditions. Harmonic functions are real analytic in the interior of their domains, so discarding $R_2(\Delta)$ gives a controlled local linearization when an effective remainder bound is available and Δ is small; it is not a long-horizon guarantee.

Busy Beaver is instead a discrete noncomputable extremal function with no computable Taylor envelope. A polynomial can interpolate any finite table of its values perfectly while saying nothing about the next value, and every total computable waiting schedule derived from a polynomial, exponential, or Taylor model is eventually exceeded. The comparison concerns effectively specified analytic laws: coefficients or boundary data can themselves encode noncomputable information.

This barrier is stronger than sensitive dependence or visual complexity alone in a precise decision-theoretic sense. A finite contractive iterated-function system or chaotic map with computable data can generate an intricate phenotype from a short rule, yet Turing universality additionally creates questions about the unbounded future for which no terminating general algorithm exists.

The categories can overlap when a dynamical system performs universal computation. This is a worst-case claim across generator descriptions, not evidence that typical financial markets exhibit Busy-Beaver delays [17, 21, 22].

These obstructions concern unbounded horizons. At any specified finite horizon, total programs can still be run and their realized prices and gains computed.

D. Worst-Case Value and Private Randomization

Returning to the direct countermarket, its finite-horizon value can be stated explicitly, and the role of a trader's hidden randomization can be separated from deterministic pathwise opposition.

Corollary 4 (Maximin Value). *At every finite horizon T, if the neutral strategy is allowed, then*

$$\sup_{\text{total computable } M_\sigma} \inf_{P \in \mathcal{P}^+_{\text{comp}}} G_T(M_\sigma, P) = 0.$$

Proof. The neutral strategy guarantees zero, while Theorem 1 gives every strategy a computable countermarket with gain at most zero. □

Proposition 5 (Private Randomization Does Not Create a Guarantee). *Fix a finite horizon T, a rational $P_0 > 0$, and a rational $\epsilon \in (0, 1)$. There is a single passive computable market law, independent of the trader, under which every predictable randomized strategy with integrable gains has expected gain zero. Specifically, let*

$$P_{t+1} = P_t(1 + \epsilon\xi_{t+1}), \qquad t = 0, \dots, T-1,$$

where $(\xi_t)_{t\geq 1}$ are independent fair signs, also independent of the trader's private random seed. If w_t is measurable with respect to the information available when the position is chosen and $w_t(P_{t+1} - P_t)$ is integrable, then

$$G_n = \sum_{t=0}^{n-1} w_t(P_{t+1} - P_t), \qquad n = 0, \dots, T,$$

is a martingale. In particular, $\mathbb{E}[G_T] = 0$.

Proof. Let R denote the trader's private seed and set $\mathcal{G}_t = \sigma(P_0, \dots, P_t, R)$. Allowing $\mathcal{G}_t$ to contain all of R only strengthens the claim. Predictability makes w_t $\mathcal{G}_t$-measurable, while ξ_{t+1} is independent of $\mathcal{G}_t$ and has mean zero. Hence

$$\mathbb{E}[w_t(P_{t+1} - P_t) \mid \mathcal{G}_t] = \epsilon P_t w_t \, \mathbb{E}[\xi_{t+1} \mid \mathcal{G}_t] = 0.$$

The integrability assumption therefore makes $(G_n)_{n=0}^T$ a martingale, and $\mathbb{E}[G_T] = \mathbb{E}[G_0] = 0$. □

Private randomization prevents a deterministic countermarket based only on the public history from mechanically opposing the unseen current draw, but it does not yield strictly positive expected gain in every environment.

a. Optional online-game reading The recursion in Theorem 1 also defines a finite perfect-information game: the trader announces w_t, the environment chooses the next return, and Corollary 4 gives the trader-first maximin value zero. This reading is optional because the fixed deterministic trader's countermarket program already generates the same path offline. Borel determinacy concerns infinite games with Borel payoff sets and does not by itself identify a winner here [23]. Hidden private randomization is a different information structure: Proposition 5 proves zero expected gain under one passive computable fair-sign law, not a mixed-game value without specified strategy spaces.

E. A Passive Market: Martin-Löf Randomness

Unlike the tailored countermarket, the path considered here is not constructed from the trader's code. Suppose first that an effective binary coding of price directions is Martin-Löf random for the fair-coin measure λ. More generally, algorithmic randomness is always assessed with respect to an explicitly specified computable reference measure. Here "random" means algorithmically random, not merely irregular-looking or sensitive to initial conditions. Intuitively, one first lists every statistically exceptional pattern that can be detected by an effective procedure and whose probability can be driven effectively to zero. A Martin-Löf-random sequence escapes every such test. It may still contain long runs, apparent trends, and many dates that a trader guesses correctly; no single finite pattern disqualifies it.

Algorithmic information theory gives a complementary description in terms of the length of the shortest explanation. Fix a universal prefix-free machine U, and define the prefix complexity of a finite string s by

$$K_U(s) = \min\{|p| : U(p) = s\}. \tag{9}$$

Changing U changes this quantity by at most an additive constant. The Levin–Schnorr characterization states, for fair-coin measure, that an infinite binary sequence x is Martin-Löf random if and only if there is a constant c such that

$$K_U(x\restriction n) \geq n - c \qquad \text{for every } n. \tag{10}$$

Thus no effective description compresses its prefixes by an unbounded number of bits: no description is shorter than the raw n-bit prefix by more than a fixed constant [24–26].

This is stronger than emergent visual complexity. A fractal or chaotic trajectory generated from a short computable rule may look elaborate while retaining the description "run this law from these data"; an algorithmically random record has no uniformly effective compression of its prefixes. The distinction is asymptotic: no finite price sample establishes Martin-Löf randomness, and K_U is uncomputable. Operationally, Busy Beaver prevents a computable deadline for abandoning every shorter program still running. Moreover, every finite word extends both to a computable nonrandom sequence and to Martin-Löf-random sequences. Failed compression is therefore not a randomness certificate, and a direction-only coding omits magnitudes, timing, costs, and other features relevant to profit.

In the fair binary market, a nonnegative capital process $K\colon \{0,1\}^{<\mathbb{N}} \to \mathbb{R}_{\geq 0}$ is a martingale when

$$K(s) = \frac{K(s0) + K(s1)}{2}. \tag{11}$$

Here $K(s)$ denotes capital after the history s, not prefix complexity. A supermartingale replaces the equality in Eq. (11) by

$$K(s) \geq \frac{K(s0) + K(s1)}{2};$$

its expected next capital does not exceed its present capital. Lower semicomputability means that every value can be approximated uniformly from below by an increasing computable sequence of rationals.

Theorem 6 (Ville's Inequality)**.** *Let $(K_n)_{n\geq 0}$ be a nonnegative supermartingale, adapted to the natural filtration under the fair-coin measure λ, with $K_0 = 1$. Then, for every $a > 0$,*

$$\lambda\left(\left\{x : \sup_{n\geq 0} K_n(x) \geq a\right\}\right) \leq \frac{1}{a}. \tag{12}$$

Ville's inequality follows by stopping the nonnegative capital process when it first crosses a and applying optional stopping, followed by a limit [27]. In particular, the set on which a fixed nonnegative supermartingale becomes unbounded has probability zero.

To state the effective refinement correctly, let λ denote fair-coin measure on Cantor space $\{0,1\}^{\mathbb{N}}$. A Martin-Löf test is a uniformly computably enumerable sequence of open sets $(U_k)_{k\geq 1}$ satisfying $\lambda(U_k) \leq 2^{-k}$. A sequence is *Martin-Löf random* if it avoids $\bigcap_k U_k$ for every such test [28]. A sequence is *computably random* if no total computable nonnegative martingale succeeds on it by becoming unbounded. A *Schnorr-random* sequence passes every Martin-Löf test whose component measures are uniformly computable. These are distinct notions, with strict implications

$$\begin{aligned} \text{Martin-Löf random} &\implies \text{computably random} \\ &\implies \text{Schnorr random.} \end{aligned} \tag{13}$$

Neither converse holds [26, 29, 30].

The distinction between these notions also matters for the trader–market interpretation. A total computable martingale is an executable idealized betting rule: on each finite history it returns its capital and wager in finite time. A lower-semicomputable supermartingale is a broader constructive test whose values can be approximated from below; it need not itself provide a deadline-bound trading action. The exact characterization of Martin-Löf randomness uses this broader class. Nevertheless, because Martin-Löf randomness implies computable randomness, it also blocks unbounded capital for every total computable nonnegative fair-game martingale.

Theorem 7 (Effective Martingale Barrier)**.** *A binary sequence x is Martin-Löf random if and only if no lower-semicomputable nonnegative supermartingale succeeds on x by unbounded capital. Consequently, the success set of each such betting strategy is effectively λ-null, and the set of paths on which all of them remain bounded has fair-coin measure one.*

Proof. Normalize a lower-semicomputable nonnegative supermartingale d by $d(\varnothing) \leq 1$, and let

$$U_k = \{x : \exists n,\ d(x{\upharpoonright}n) > 2^k\}.$$

Lower semicomputability makes U_k computably enumerable and open, while Ville's inequality gives $\lambda(U_k) \leq 2^{-k}$. Hence (U_k) is a Martin-Löf test and no Martin-Löf-random sequence permits d to become unbounded. Conversely, every Martin-Löf test can be converted into a lower-semicomputable supermartingale that succeeds on its intersection; this is the martingale characterization of Martin-Löf randomness [25, 26]. □

The same construction extends from λ to any computable probability measure μ. In cylinder notation the supermartingale condition becomes

$$d(s)\mu([s]) \geq d(s0)\mu([s0]) + d(s1)\mu([s1]).$$

Remark 2 (Computability Versus an ELMM). A computable reference measure μ supplies effective cylinder probabilities; computability alone is not a financial no-arbitrage condition. For μ to be an equivalent local martingale measure, it must be defined on the same filtered model as the physical measure $\mathbb{P}$, satisfy $\mu \sim \mathbb{P}$, and make the discounted traded-price process a local μ-martingale. None of these properties follows from computability, and conversely an ELMM need not be computable. Likewise, an abstract μ-test supermartingale is not automatically the gain process of an admissible self-financing portfolio; that interpretation requires traded payoffs and financing constraints that realize the corresponding bets.

Theorem 7 is asymptotic and reference-measure relative: it blocks unbounded test capital on a measure-one class, not every finite-horizon loss, and it does not extend unchanged to unrestricted borrowing or a non-martingale physical law.

Corollary 8 (No Sustained Computable Prediction Edge)**.** *Let $x = (x_t)_{t\geq 1} \in \{0,1\}^{\mathbb{N}}$ be Martin-Löf random for the fair-coin measure, and let $F\colon \{0,1\}^{<\mathbb{N}} \to \{0,1\}$ be a total computable next-bit predictor. Define*

$$\widehat{x}_t = F(x_1 \cdots x_{t-1}), \qquad A_T = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{\widehat{x}_t = x_t\}.$$

Then

$$\liminf_{T\to\infty} A_T \le \frac{1}{2}.$$

Thus no total computable predictor has a fixed positive asymptotic accuracy advantage on a fair-coin Martin-Löf-random path.

Proof. Suppose instead that $\liminf_{T\to\infty} A_T > 1/2$. Choose $\eta > 0$ and T_0 such that

$$A_T \ge \frac{1}{2} + \eta \qquad \text{for all } T \ge T_0,$$

and choose a rational δ satisfying $0 < \delta < \min\{1, 2\eta\}$. Starting from $K_0 = 1$, bet the fraction δ of current capital on the bit predicted by F. Thus

$$K_t = \begin{cases} K_{t-1}(1+\delta), & \widehat{x}_t = x_t, \\ K_{t-1}(1-\delta), & \widehat{x}_t \ne x_t. \end{cases}$$

Exactly one of the two possible next bits agrees with the prediction, so this recursion defines a total computable nonnegative fair-coin martingale. Along x,

$$\begin{aligned} \frac{1}{T}\log K_T &= A_T \log(1+\delta) + (1-A_T)\log(1-\delta) \\ &\ge g_\eta(\delta), \\ g_\eta(\delta) &= \left(\frac{1}{2}+\eta\right)\log(1+\delta) \\ &\quad + \left(\frac{1}{2}-\eta\right)\log(1-\delta) \end{aligned}$$

for every $T \ge T_0$. Now $g_\eta(0) = 0$ and

$$g'_\eta(\delta) = \frac{2\eta - \delta}{1-\delta^2} > 0 \qquad \text{when } 0 < \delta < \min\{1, 2\eta\}.$$

Hence $g_\eta(\delta) > 0$, and K_T grows exponentially along x, contradicting Theorem 7. Therefore the asserted sustained advantage is impossible. □

This corollary concerns binary prediction under the fair-coin reference law. It does not by itself imply trading profit or loss when returns have unequal magnitudes, financing constraints, or transaction costs; finite streaks and isolated correct forecasts remain possible.

A total deterministic computable generator cannot itself output a fair-coin Martin-Löf-random sequence. The countermarket is correspondingly computable and tailored to one trader, whereas a random record is passive and typical relative to its stated measure. A hybrid can apply computable dynamics to genuinely random input, but the randomness then comes from that input. Game-theoretic probability develops the associated pathwise betting viewpoint systematically [31].

F. Ramsey Regularity and the Finite-Pattern Trap

A large market record can be pattern-rich even when it contains no exploitable law. Ramsey theory makes this finite combinatorial point without assuming a probability model. For fixed positive integers c, m, q, there is a finite number $R_q(m; c)$ such that every c-coloring of the q-element subsets of any set of size at least $R_q(m; c)$ has an m-element subset whose q-subsets all receive the same color. Thus every red–blue coloring of the pairs among six objects contains a monochromatic triangle. Van der Waerden's theorem similarly forces a monochromatic arithmetic progression of any prescribed finite length in every sufficiently long finite coloring [32, 33]. These thresholds may be enormous, but they are finite and computable, unlike a Busy-Beaver horizon.

The guarantee depends on the coding and is weaker than a trading signal. In a discretized return record, equal symbols at equally spaced dates need not be consecutive, form a price trend, have a favorable sign, or persist beyond the sample. Ramsey existence therefore asserts neither statistical dependence nor causation, predictability, or profit. Nor does it conflict with algorithmic randomness. A fair-coin Martin-Löf-random sequence is Borel normal, so every finite binary word, including arbitrarily long runs and finite chart motifs, occurs with its fair-coin limiting frequency. Randomness here means escaping every effective null test and lacking a global effective compression or successful test-capital process, not local patternlessness [25, 26].

Calude and Longo call regularities forced by data volume *Ramsey-type correlations*. Their incompressibility argument emphasizes that almost all sufficiently long finite strings resist compression by any fixed positive fraction while still containing the finite regularities forced by Ramsey theory [34]. This is an epistemic warning, not a theorem that every sample correlation is false: a Ramsey configuration is not itself a correlation coefficient, and a pattern observed in one finite history is compatible with both random and computable continuations.

Trading research compounds this pattern abundance with selection. If all M null hypotheses are true, their rejection events are independent, and each has rejection probability exactly α, the probability of at least one false rejection is $1-(1-\alpha)^M$; if their probabilities are at most α, this is an upper bound. Without independence, the union bound gives $\min\{1, M\alpha\}$ as an upper bound. This probabilistic multiple-testing problem is distinct from Ramsey's deterministic existence theorem, but its practical lesson is aligned: a trend or cross-market relation retained after searching many assets, windows, transformations, lags, and parameters is a candidate hypothesis, not yet an edge. A defensible FAPP claim must report the search universe and survive selection correction, costs, a genuinely untouched holdout or independent replication, and testing outside the selected regime [35].

G. What Rice's Theorem Does and Does Not Say

Rice's theorem makes every nontrivial extensional property of arbitrary partial computable functions undecidable [36]; in particular, totality is undecidable in general [12]. It does not make a fixed total strategy's gain on a completed finite computable path undecidable, because that program can simply be simulated. Theorems 2 and 3 instead give the required promise-domain results by explicit Halting reductions: the unbounded questions of eventual profit and future-event occurrence are undecidable even though every code produced by the reductions is total.

III. THE NO-ARBITRAGE FOUNDATION

Here the market is no longer an opponent choosing a response after seeing the trader's action. Instead, financial admissibility and no-arbitrage delimit which trader–market pairs are coherent.

A. Discounted Gains and Admissibility

Let the market be defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in[0,T]}, \mathbb{P})$. Let $B_t > 0$ be a numeraire and S_t the vector of traded asset prices. Write $\overline{S}_t = S_t/B_t$ for discounted prices. A predictable, $\overline{S}$-integrable strategy H has discounted self-financing gain

$$G_t = (H \cdot \overline{S})_t, \qquad G_0 = 0. \tag{14}$$

It is *admissible* if there is a finite constant a such that

$$G_t \geq -a \quad \text{for all } t \in [0,T], \quad \mathbb{P}\text{-a.s.} \tag{15}$$

This uniform lower bound rules out doubling systems financed by unbounded credit.

Theorem 9 (Universality Implies Classical Arbitrage)**.** *If an admissible self-financing strategy H with zero initial capital satisfies*

$$G_T \geq 0 \quad \mathbb{P}\text{-a.s.}, \qquad \mathbb{P}(G_T > 0) > 0,$$

then H is a classical arbitrage. In particular, a pointwise universal strategy is a classical arbitrage whenever its gain is defined on an admissible path class carrying $\mathbb{P}$.

Proof. The displayed payoff conditions, together with zero initial cost, self-financing, and admissibility, are the standard classical-arbitrage conditions. Pointwise strict positivity on every admissible path implies them, with $\mathbb{P}(G_T > 0) = 1$. □

Remark 3 (Terminology and Strength). The pathwise condition is sometimes called a sure or strong arbitrage, but it is not synonymous with *arbitrage of the first kind*. The latter is the condition dual to No Unbounded Profit with Bounded Risk (NUPBR/NA1) and is a distinct, weaker viability concept [37]. Classical no-arbitrage alone already excludes the universal strategy in Theorem 9; NFLVR is invoked below because it supplies the standard martingale-measure formulation.

B. NA, NUPBR, and Benchmark-Relative Gains

The relevant no-arbitrage notions do not form the simple chain sometimes suggested in informal discussions. In the standard semimartingale setting, NFLVR combines classical NA with NUPBR, whereas NA and NUPBR are in general logically distinct. NUPBR is equivalent, under standard portfolio hypotheses, to the existence of a numeraire portfolio whose positive relative-wealth processes are supermartingales [37, 38]. It may hold even when the stronger NFLVR condition and an ELMM fail.

This distinction prevents two overstatements. First, Theorem 9 uses only classical NA; NUPBR alone is not silently substituted for it. If a zero-cost gain were nonnegative at all intermediate times and positive at terminal time, scaling would violate NUPBR, but terminal pathwise positivity together with only a strategy-specific lower bound does not by itself supply that scaling argument. Second, benchmark and stochastic-portfolio approaches can generate gains *relative* to a numeraire or market portfolio without producing zero-cost positive gain on every path. Relative arbitrage in diverse equity markets and real-world benchmark pricing therefore do not contradict the theorem [39, 40].

C. The Martingale Constraint

For a locally bounded semimartingale price process, the Delbaen–Schachermayer form of the First Fundamental Theorem states that No Free Lunch with Vanishing Risk (NFLVR) is equivalent to the existence of an equivalent local martingale measure (ELMM) $\mathbb{Q} \sim \mathbb{P}$ under which $\overline{S}$ is a local martingale [41, 42]. For general unbounded semimartingales the corresponding formulation uses an equivalent σ-martingale measure [43]; the corollary below retains the locally bounded setting.

Corollary 10 (No Universal Zero-Cost Strategy)**.** *If an ELMM $\mathbb{Q}$ exists, no admissible self-financing strategy with zero initial capital can satisfy $G_T \geq 0$, $\mathbb{P}$-almost surely, and $\mathbb{P}(G_T > 0) > 0$. Consequently, no pointwise universal strategy exists.*

Proof. Under $\mathbb{Q}$, the admissible stochastic integral $G = H \cdot \overline{S}$ is a local martingale bounded from below and therefore a supermartingale. Hence $\mathbb{E}_{\mathbb{Q}}[G_T] \leq G_0 = 0$. If the displayed arbitrage conditions held, equivalence would give $G_T \geq 0$, $\mathbb{Q}$-almost surely, with $\mathbb{Q}(G_T > 0) > 0$. Consequently $\mathbb{E}_{\mathbb{Q}}[G_T] > 0$, a contradiction. □

The ELMM conclusion concerns discounted gains under the pricing measure $\mathbb{Q}$. It does not say that every risky strategy has non-positive expected return under the physical measure $\mathbb{P}$; positive physical expected returns may compensate risk.

D. Win Rate Is Not Expected Value

The probability of closing a trade at a profit can greatly exceed $1/2$ even in a fair market. Optional stopping gives the precise distinction.

Proposition 11 (Asymmetric Barriers in a Fair Random Walk)**.** *Let X_n be a simple symmetric random walk with $X_0 = 0$, and for positive integers a, b let*

$$\tau = \inf\{n \geq 0 : X_n = a \text{ or } X_n = -b\}.$$

Then

$$\mathbb{P}(X_\tau = a) = \frac{b}{a+b}, \qquad \mathbb{E}[X_\tau] = 0. \tag{16}$$

Proof. For this finite interval, $\tau < \infty$ almost surely and $|X_{n\wedge\tau}| \leq \max\{a, b\}$. Optional stopping gives $\mathbb{E}[X_{n\wedge\tau}] = 0$, and bounded convergence yields $\mathbb{E}[X_\tau] = 0$ [44]. If $p = \mathbb{P}(X_\tau = a)$, then $0 = \mathbb{E}[X_\tau] = ap - b(1-p)$, which gives $p = b/(a+b)$. □

With a take-profit of $a = 1$ and a stop-loss of $b = 9$, the win probability is 0.9, but the expected payoff is zero: nine frequent unit gains are offset by one nine-unit loss in expectation. Thus prediction accuracy, trade win rate, and expected discounted profit are three different quantities.

IV. THE COMBINATORIAL PERSPECTIVE

Here the opposing side is an ensemble of market records rather than an adaptive market. The optimization NFL theorem and the elementary prediction result needed for market-direction claims should be distinguished.

Theorem 12 (Optimization NFL, Wolpert–Macready)**.** *Let X and Y be finite and give the function class Y^X its uniform law. For any two non-revisiting black-box algorithms and any fixed number $q \leq |X|$ of queries, the resulting q-term sequences of observed values have the same distribution. Consequently, every performance functional depending only on that observed value sequence has the same average for both algorithms [45].*

The theorem concerns oracle-based optimization over function spaces. It does not, by itself, imply a statement about self-financing gains on stochastic price paths. A sharpening makes its structural assumption explicit: for a uniform distribution on a finite function class, the NFL equality for all non-revisiting algorithms holds precisely when the class is closed under permutations of the search domain [46]. Financial path classes are generally not permutation-closed: temporal order, continuity, financing, and no-arbitrage restrictions distinguish one permutation from another. Thus a financial edge, when present, resides in exactly such symmetry-breaking structure and the inductive bias used to model it.

For binary directional prediction the relevant result is simpler and can be derived directly.

Proposition 13 (Uniform Binary Prediction)**.** *Let $X_1, \ldots, X_T$ be uniformly distributed on $\{0,1\}^T$. A causal deterministic predictor chooses $\widehat{X}_t = f_t(X_1, \ldots, X_{t-1})$. Its mean accuracy*

$$A_T = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{\widehat{X}_t = X_t\}$$

satisfies $\mathbb{E}[A_T] = 1/2$. The same holds for a randomized predictor driven by a private random seed independent of $(X_1, \ldots, X_T)$.

Proof. Conditional on every history $X_1, \ldots, X_{t-1}$, the next bit X_t is an independent fair bit. Therefore $\mathbb{P}(\widehat{X}_t = X_t \mid X_1, \ldots, X_{t-1}) = 1/2$. Linearity of expectation gives the result. In the randomized case, conditioning additionally on the independent private seed leaves X_t fair given the history and proves the same equality. □

Consequently, no predictor can achieve expected directional accuracy greater than $1/2$ under the uniform distribution over all binary sequences; in particular, none can exceed $1/2$ on every sequence. This is an NFL symmetry statement, not a model of actual markets. Financial sign sequences are not uniform in general, and a directional forecast does not determine trading profit because payoff sizes, financing, and stopping rules matter (Proposition 11).

The defensible practical conclusion is conditional: a claim of repeatable success must identify non-uniform structure in the physical measure $\mathbb{P}$, such as a trend, mean reversion, a risk premium, information, or an institutional constraint that breaks the NFL symmetry. The hypothesized edge concerns a restricted environment, not universal algorithmic superiority.

V. A PRACTICAL ROBUSTNESS DIAGNOSTIC: TIME REVERSAL

This section turns the trader–market leitmotif into a falsification tool. Hold the trader fixed and transform only the market record. Among the countable family of computably specified stress transformations, time reversal is one of the simplest, particularly for detecting dependence on directional persistence. It is not a further impossibility theorem on the level of the Halting reductions or no-arbitrage.

Let $\mathcal{M}$ be a space of strictly positive paths $P\colon [0,T] \to \mathbb{R}^n_{>0}$. The time reversal of P is

$$\widetilde{P}(t) = P(T-t), \qquad t \in [0,T]. \tag{17}$$

Literal reversal reuses the observed price levels and preserves the magnitudes of log-price moves while reversing their order and signs. It is not the same as merely negating arithmetic returns: if a forward gross return is R, the corresponding reversed gross return is $1/R$. In an empirical implementation the strategy must be rerun causally from scratch on the reversed record; reversing the original orders or realized profit would leak future information. For a causal strategy σ, let $\Pi_t[\sigma; P]$ denote its discounted gain from zero initial capital through time t on P. Call σ *pathwise admissible* on $\mathcal{M}$ if some finite C satisfies $\Pi_t[\sigma; P] \geq -C$ for every $P \in \mathcal{M}$ and $t \in [0, T]$.

Definition 1 (Pathwise Universal Strategy)**.** A causal strategy σ is *pathwise universal* on $\mathcal{M}$ if it is self-financing and pathwise admissible there and satisfies $\Pi_T[\sigma; P] > 0$ for every $P \in \mathcal{M}$.

Criterion 1 (Reversal Falsification Test)**.** Suppose $\mathcal{M}$ is closed under time reversal. If there exists $P \in \mathcal{M}$ such that

$$\min\{\Pi_T[\sigma; P], \Pi_T[\sigma; \widetilde{P}]\} \leq 0, \tag{18}$$

then σ is not pathwise universal on $\mathcal{M}$.

Proof. Both P and $\widetilde{P}$ belong to $\mathcal{M}$. A non-positive gain on either member of the pair contradicts Definition 1. □

One non-winning path refutes a claim quantified over every path. Reversal supplies a paired candidate, not a proof that one member must fail: a causal strategy may take different positions on the reversed history. Diagonalization and no-arbitrage provide the genuine impossibility results under their respective assumptions.

Remark 4 (Computable Stress Tests Beyond Reversal). Time reversal is one of countably many computable probes: sign reversal, regime splicing, jumps, delays, volatility rescaling, financing, and transaction costs. Since program totality is undecidable, an operational battery must select known-terminating transformations whose outputs remain in the claimed market class. Here "test" means empirical stress, not a Martin-Löf test: a uniformly computably enumerable sequence of open sets with effective measure bounds. Reversal defines no null set and certifies no randomness; passing it shows only tested robustness, not universality.

Remark 5 (Filtrations and Time Reversal). Stochastic model classes are not automatically reversal-closed: $P(T-t)$ is generally not adapted to the forward filtration, and reversing a diffusion requires additional hypotheses [47]. For example, if

$$X_t = \log(S_t/S_0) = mt + \sigma W_t,$$

then $X_{T-t} - X_T$, under its reversed filtration, has drift $-m$ and volatility σ, so a class containing both drift signs is closed in law under reversed increments. Physically, $m = \mu - \sigma^2/2$; under a risk-neutral measure the discounted stock may be a martingale while its logarithm retains Itô drift $-\sigma^2/2$. Neither an ELMM nor a driftless arithmetic-Brownian special case establishes detailed balance in general. The diagnostic remains pathwise; Section III gives the probabilistic constraints.

VI. A BRIEF SOCIAL-CHOICE ANALOGY

Arrow's Impossibility Theorem and the trading result share only an unrestricted-domain structure. With at least three alternatives, unrestricted preference profiles, collective rationality, Pareto unanimity, and independence of irrelevant alternatives force a social-welfare aggregation rule to be dictatorial [48]. Arrow quantifies over preference profiles; pathwise universal trading quantifies over price paths. Neither theorem implies the other, but both show that natural requirements can become jointly untenable when imposed on every admissible input. In the fixed-point language surveyed by Yanofsky [49], Eq. (3) anti-aligns the return with every nonzero position while leaving the neutral response $w_t = 0$; that fixed point yields zero gain, not strict profit.

VII. CASE STUDY: THE WHEEL OPTIONS STRATEGY

The Wheel makes the trader–market configuration concrete. It sells a cash-secured put and, after assignment, sells a covered call against the acquired stock. Ignoring dividends and financing in the terminal-payoff identity, put–call parity gives

$$S_T - (S_T - K)^+ = K - (K - S_T)^+ = \min\{S_T, K\}. \tag{19}$$

After the corresponding time-zero costs are matched, a covered call and a cash-secured put therefore have the same European terminal payoff. Rolling the Wheel repeatedly renews a put-like, short-downside-convexity exposure; it does not manufacture directionless income.

Option selling can nevertheless have positive mean return under a physical law. The difference between risk-neutral and physical expectations of future variance is commonly called the variance risk premium, with sign conventions varying across the literature, and option-market evidence documents compensation for bearing volatility and jump risk [50, 51]. Such compensation is empirical and state-dependent: its sign and magnitude depend on option prices, strikes, transaction costs, and the physical distribution.

Equation (19) exposes four practical failure regimes. A crash after assignment produces essentially one-for-one downside below K. A persistent decline can make repeated premiums too small to offset the renewed short-put exposure. In a large rally, assignment and call exercise can still leave a positive absolute profit, but the cap at K can underperform a buy-and-hold benchmark. Finally, capital exhaustion is absorbing for a self-financing trader: if $W_{t+1} = W_t(1 + fR_{t+1})$ and $1 + fR_{t+1} = 0$, then

$W_{t+1} = 0$ unless external capital is injected. A fully cash-secured, unlevered implementation has bounded loss, but an underlying that falls to zero can consume almost all committed capital net of premium.

A rally/crash reversal is therefore a useful paired stress test, although the payoff identity, rather than reversal alone, supplies the economic mechanism. Repeated small premiums also need not imply satisfactory time-average growth:

$$\frac{1}{T}\log\frac{W_T}{W_0} = \frac{1}{T}\sum_{t=1}^{T}\log(1+fR_t),$$

so frequent gains can coexist with poor long-run growth when rare losses are large [52]. The Wheel may be a defensible FAPP strategy under a specified risk premium, capital policy, cost model, and benchmark; it is not a pathwise profit guarantee.

VIII. UNIVERSAL PORTFOLIOS AND DISTANCE FROM UNIVERSALITY

Here the trader changes the claim made against the market side: it seeks small regret to a benchmark portfolio class, not positive profit on every path. The word "universal" also appears in a celebrated positive theorem. For price-relative vectors $x_t \in \mathbb{R}^m_{>0}$, a constant-rebalanced portfolio b has wealth

$$S_T(b) = \prod_{t=1}^{T} b^\top x_t.$$

Cover's universal portfolio is nonanticipating and, under the conditions of his theorem, has the same asymptotic exponential growth rate as the best constant-rebalanced portfolio chosen in hindsight on every individual bounded sequence [53]. In regret notation,

$$\frac{1}{T}\left[\log\max_b S_T(b) - \log S_T^{\mathrm{UP}}\right] \longrightarrow 0. \tag{20}$$

Equation (20) is a relative guarantee, not an absolute-profit guarantee. If every constant-rebalanced portfolio loses money on a path, the universal portfolio may lose as well while retaining vanishing per-period regret. Cover's result therefore does not contradict Corollary 10; it illustrates a useful benchmark-relative meaning of "universal" once the comparison class is specified.

Online regret methods also yield financial guarantees of another kind. DeMarzo, Kremer, and Mansour show that gradient strategies designed to minimize asymptotic regret imply trading strategies that provide arbitrage-based option price bounds without assuming continuous price paths, complete markets, or a pricing kernel; the bounds depend on realized quadratic variation [54]. This is robust pricing and hedging, not guaranteed positive profit or next-move prediction.

a. A Hierarchy of Attainable Claims The useful alternatives to pathwise universality are not weaker points on one single order; they change the benchmark, horizon, or probability quantifier. Table II records the distinctions.

Superhedging duality gives another quantitative notion: once a contingent claim and an admissible model class are fixed, its superhedging price is the least initial capital needed for a pathwise guarantee. Requiring zero initial capital while demanding a strictly positive payoff is precisely the degenerate, arbitrage-producing endpoint ruled out above.

IX. FAPP STRATEGIES AND THEIR HONEST SCOPE

Although pathwise universal profit is excluded, a strategy may be profitable *for all practical purposes* (FAPP, following Bell [55]) relative to a physical law $\mathbb{P}$, a benchmark, and a finite horizon. Market making may earn spreads when adverse selection and inventory risk are controlled, and Kelly investing maximizes expected logarithmic growth when the relevant distribution is sufficiently well specified [56]. Neither is an unconditional guarantee.

Here "insight" means a falsifiable restriction on the market side. In actual trading it usually appears as a heuristic: persistence, mean reversion, valuation correction, or compensation for an identified risk. The fact that a heuristic is algorithmic does not disqualify it; its edge is a regime-dependent empirical hypothesis. A repeatable ex ante claim must therefore specify its market class, horizon, costs, benchmark, and validation protocol.

An informational advantage must be distinguished from prohibited insider trading. Public disclosures, proprietary analysis, and lawfully acquired data may generate beliefs not yet reflected in prices. Trading on information whose acquisition or use is forbidden by applicable insider-trading or market-abuse law lies outside the lawful practical strategy class considered here. Because legal definitions vary by jurisdiction, the mathematical adjective "informed" establishes neither legality nor illegality; even superior lawful information remains short of a pathwise guarantee.

The unit of performance matters as well. For a funded portfolio, let R^{nom} be nominal total return and π inflation over the same interval under a stated price index. Then

$$1 + R^{\mathrm{real}} = \frac{1+R^{\mathrm{nom}}}{1+\pi}. \tag{21}$$

A positive nominal return can therefore be a purchasing-power loss. The price index is an economic deflator, not necessarily a tradable numeraire for the FTAP; the consumption benchmark and financing account serve different purposes.

Persistent productivity growth is one possible restriction supporting positive real drift. Applied scientific and technological progress can raise real output [57]; if the

TABLE II. Different meanings of a successful or universal strategy.

Notion	**Guarantee**	**Status**
Pathwise universal profit	Positive discounted zero-cost gain on every admissible path	Excluded on any model class supporting classical NA
Relative arbitrage	Outperformance of a specified market or numeraire portfolio	Possible under structural market assumptions
Cover universality	Vanishing per-period log regret to the best constant-rebalanced portfolio	Pathwise but benchmark-relative
FAPP or statistical edge	Positive performance under a specified physical law, data protocol, and horizon	Conditional and empirically falsifiable

gains persist and reach listed firms as after-cost cash flows in a stable institutional regime, they may support positive expected long-horizon real total return on diversified productive capital. This is not an arbitrage or a universal law: positive capital is invested, expectations may already be priced, gains may accrue to consumers, labor, entrants, or unlisted firms, discount rates change, and innovation can displace incumbents [58]. Aggregate real growth therefore establishes neither next-move prediction nor active alpha.

Equilibrium theory likewise permits only conditional edges. Grossman and Stiglitz show that perfectly informationally efficient markets are inconsistent with costly information acquisition: if prices revealed all information without reward, agents would not pay to produce it [59]. Conversely, under its common-prior, common-knowledge, risk-aversion, and Pareto-optimality assumptions, the Milgrom–Stokey no-trade theorem limits trade based on private information alone [60]. A complementary market-selection result is also conditional: among agents in Sandroni's complete-market model who share an intertemporal discount factor and choose saving endogenously, accurate forecasters prosper and inaccurate ones are driven out [61]. These equilibrium results depend on their model assumptions; none is a pathwise trading guarantee or a substitute for the FTAP.

Deployment can also change the market side. A large, crowded, or public strategy consumes liquidity, moves quotes, attracts imitation or opposition, and changes the distribution on which it was estimated. Lo's Adaptive Markets Hypothesis provides this evolutionary interpretation [62]: competition and impact may erode an edge without producing the tailored countermarket of Theorem 1. Feedback invalidates scale-free extrapolation; it does not imply that every large strategy loses.

Finally, the finite-pattern trap of Section II F becomes operational in strategy discovery. Trying many specifications and retaining the best backtest turns its Sharpe ratio into an order statistic. The Deflated Sharpe Ratio corrects for selection, non-normality, and the number of trials [35]; untouched validation, costs, and deployment analysis remain necessary. These controls do not prove an edge, but make a finite-sample FAPP claim more honestly falsifiable.

X. CONCLUSION

Everything returns to the same pair: a trader and a market. For every fixed total deterministic computable trader, Theorem 1 constructs a fixed positive computable price path whose next bounded proportional move opposes the trader's position. The quantifiers are $\forall\sigma\,\exists\mathcal{E}_\sigma$: no physical market must observe the deployed orders, the always-flat trader earns zero, and every nonzero position loses on its trader-relative countermarket. This direct diagonal result uses no Halting-Problem reduction and identifies no single path that defeats every trader.

The remaining limits concern different market sides. Gold precludes identifying every computable binary generator in the limit, while next-bit diagonalization defeats each fixed computable predictor on its predictor-relative record. Halting reductions preclude uniform certification of eventual profit or encoded future events, and Busy Beaver excludes a computable description-size waiting horizon despite finitely solvable low-complexity cases. Under a stated computable reference measure, Martin-Löf randomness blocks unbounded success by effective test-capital processes. Ramsey regularity explains why finite structured subsequences nevertheless remain abundant, including in random records. No-arbitrage and uniform binary averaging supply still different boundaries. Their conclusions are complementary because their inputs, quantifiers, and success criteria differ.

Positive results arise only after restricting the claim. The Wheel may earn a risk premium while retaining put-like downside; Cover guarantees asymptotic performance relative to a benchmark class; and FAPP strategies may exploit a specified regime, lawful information, or compensated risk. Such performance must be evaluated after costs and financing, against an explicit benchmark and, where relevant, in purchasing-power terms. Productivity-supported real growth is one possible regime hypothesis, not a next-move predictor or a guarantee of active alpha.

Any repeatable-success claim must therefore state its market restriction, probability law, benchmark, admissibility and financing conditions, horizon, data protocol, and deployment effects. Time reversal is one robustness probe among many. Useful strategies may exist, but no total deterministic computable self-financing rule guarantees strict gain on every positive computable price path; under an arbitrage-free stochastic model, no admissible zero-cost rule delivers nonnegative discounted gain almost surely and positive gain with positive probability.

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OpenAI Codex (GPT-5) was used to assist with literature organization, LaTeX editing, and checks of derivations. The author specified the scientific direction and prompts; model output retained in the manuscript was checked against the cited sources and explicit calculations. The author accepts full responsibility for the final text.

DATA AVAILABILITY

This is a purely mathematical work, and no data or software were created or analyzed in this study.

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